CircuitBench: Benchmarking Vision-Language Models for Electronic Circuit Reasoning and Fault Diagnosis

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Abstract—Vision-language models (VLMs) are increasingly proposed for use in engineering education, including as conversational laboratory assistants capable of interpreting circuit diagrams and diagnosing faults. Before such systems are deployed, their reliability on structured electrical-engineering tasks must be characterized. We present CircuitBench, a source-backed benchmark for evaluating VLMs on electronic-circuit understanding, reasoning, and fault diagnosis, derived from two publicly available datasets: the AITEE Electrical Circuit Dataset (828 annotated schematic images [1]) and the CGHD hand-drawn circuit dataset [2]. The benchmark comprises 8,598 items across three task categories: component recognition (Task A), connectivity and topology reasoning (Task B), and fault diagnosis (Task E), with a 2,581-item held-out test split. We evaluate three locally-hosted open-weight VLMs under a standardized zero-shot prompting protocol on a development-split sample. The best-performing model, Gemma 3 4B, achieves 38.3% overall accuracy (n=81), with markedly higher performance on binary yes/no subtasks (fault presence: 87.5%; topology structure: 87.5%) than on counting or free-text tasks (node count: 0%; circuit description: 0%). LLaVA 7B achieves 23.3% and LLaVA-Phi3 achieves 8.0%. All models fail completely on connectivity-counting and free-text description tasks. These results suggest that current small open-weight VLMs are not sufficiently reliable for autonomous use in electronics education or circuit debugging, and that fault detection and topology reasoning present distinct capability profiles worth investigating further.

Index Terms—vision-language models, multimodal reasoning, electrical circuits, fault diagnosis, engineering education, benchmark

I. Introduction

Vision-language models have demonstrated strong performance on visual question answering, scientific diagram interpretation, and chart understanding [3]–[5]. As these models improve, there is growing interest in applying them to engineering education—for example, as conversational laboratory assistants that can identify circuit components, trace signal paths, and diagnose wiring errors [6].

However, circuits are not generic images. Interpreting an electrical schematic requires recognizing specific symbols (resistors, voltage sources, grounds), understanding how those symbols are connected by wires and nodes, reasoning about series and parallel topologies, and—for fault diagnosis—identifying which connections or components violate expected circuit structure. An incorrect diagnosis from an AI assistant could mislead a student attempting to debug a real circuit.

Before such systems are deployed, it is important to characterize whether contemporary VLMs can perform these tasks reliably. Prior benchmarks focus on natural images [3], scientific figures [4], and charts [5], but systematic evaluation on electrical circuit diagrams has received limited attention. The AITEE project introduced annotated educational circuit images and an agentic tutoring pipeline [1], [6] but did not conduct a cross-model VLM benchmark. The CGHD dataset provides rich annotations for hand-drawn circuits [2], [7] but is a recognition resource, not an evaluation benchmark.

We introduce CircuitBench, a source-backed benchmark that fills this gap. It spans three circuit-reasoning task categories, is derived from two publicly licensed datasets, and includes controlled fault-injection items with deterministic ground truth. We evaluate three open-weight VLMs under identical zero-shot conditions and analyze their failure modes.

Contributions:

- CircuitBench, a benchmark of 8,598 items derived from AITEE and CGHD with full provenance metadata and a frozen 2,581-item test split;
- a controlled fault-injection protocol producing 798 fault-injected images with deterministic labels from 200 source circuits;
- an empirical comparison of three open-weight VLMs under standardized zero-shot prompting; and
- an error analysis identifying recurring failure modes across models and task categories.

II. Related Work

A. Vision-Language Models and Multimodal Reasoning

Large vision-language models such as GPT-4o [8], Gemini [9], and LLaVA [10] combine a visual encoder with a language model decoder trained on image-text pairs. MMBench [3] and ScienceQA [4] provide broad multimodal evaluation, while ChartQA [5] focuses on chart comprehension. Structured technical diagrams such as circuit schematics differ substantially from the natural images and charts that dominate these benchmarks, and their evaluation remains underexplored.

B. AI for Engineering Education

AI tutoring systems have been applied across STEM subjects, with vision-language models enabling diagram-based question answering [4]. The AITEE system [6] is the most directly relevant recent work: it introduces 828 annotated first-semester circuit images and proposes an agentic tutoring pipeline using image-to-netlist conversion. CircuitBench uses AITEE as its primary data source.

C. Circuit Diagram Recognition and Understanding

Automated circuit interpretation has been studied as a pipeline: detect symbols, extract connectivity, generate a netlist. Thoma et al. [7] introduced the CGHD dataset and demonstrated deep-learning-based component localization in hand-drawn circuit images. Alakhras et al. [11] showed that combining deep learning, connected-component labeling, and OCR achieves 96.67% netlist generation accuracy on digital schematics. Nguyen et al. [12] applied Graph Attention Networks to netlist parsing of integrated-circuit images. These works extract structured representations from circuit images; CircuitBench instead evaluates whether VLMs can answer natural-language reasoning questions without access to extracted annotations.

D. Technical Visual-Reasoning Benchmarks

ScienceQA [4] and ChartQA [5] evaluate multimodal reasoning on science figures and charts, respectively, but neither includes electrical schematics or circuit-level reasoning tasks. To our knowledge, CircuitBench is the first benchmark specifically designed to evaluate VLMs on component recognition, topology reasoning, and fault diagnosis from electrical circuit diagrams.

III. CircuitBench

A. Source Datasets

AITEE. The AITEE Electrical Circuit Dataset [1] contains 828 annotated schematic images of first-semester electrical engineering circuits, including series, parallel, mixed series-parallel, voltage-divider, Wheatstone bridge, delta, and star topologies. Images are accompanied by YOLO-format annotations covering 11 classes (edge/junction, ground, current source, node point, resistor, voltage source, and identifier classes). The dataset is released under the Apache 2.0 license.

CGHD. The CGHD dataset (Version 16, DOI: 10.5281/zenodo.17469897) [2] provides 3,269 annotated hand-drawn circuit diagram images contributed by 31 main drafters, plus additional real-world examination images. It includes 248,020 bounding-box annotations across 59 object classes. CGHD Version 16 is released under CC BY 4.0.

B. Data Selection

AITEE images were included if the corresponding label file contained at least one resistor annotation, ensuring non-trivial circuit content. All 828 available images met this criterion. For fault injection, images with at least three resistors and three edge/junction annotations were selected to ensure structural meaningfulness after modification; this yielded 200 source circuits.

C. Task Taxonomy

CircuitBench evaluates three task categories:

Task A — Component recognition covers three subtasks: component_presence (binary: is a given component type present?), component_count (integer: how many of a given type?), and component_enumeration (free-text: list all types with counts).

Task B — Connectivity and topology reasoning covers three subtasks: topology_structure (binary: are two components in series or parallel?), node_count (integer: how many circuit nodes?), and component_total (integer: total component count).

Task E — Fault diagnosis covers three subtasks: fault_presence (binary: is a fault present?), fault_diagnosis (free-text: identify and describe the fault), and circuit_description (free-text: describe all component types and general topology).

D. Controlled Fault Generation

Task E uses modified circuit images in which a single fault has been deterministically introduced by one of two operations:

Open circuit: a white rectangle is drawn over the wire segment between two nearby edge/junction nodes, creating a broken connection.

Disconnected component: a randomly selected resistor's bounding box (plus a 5-pixel margin) is filled white, removing the component.

Only modified copies are used; source images are never altered. Each fault produces both a presence (E1) and a diagnosis (E2) item. All transformation details are recorded in the benchmark manifest.

E. Ground-Truth Construction

Binary and integer ground truth for Tasks A and B is derived deterministically from AITEE's YOLO annotations. Fault ground truth is specified by construction: the fault type is known exactly from the transformation applied. Free-text responses are scored by keyword matching, described in Section IV.

F. Benchmark Statistics and Splits

Table I summarizes the benchmark. Items are split 70/30 test/dev by source image, keeping all items from the same source circuit in the same split to prevent leakage.

G. Licensing and Attribution

AITEE images are used under the Apache 2.0 license. CGHD images are used under CC BY 4.0. CircuitBench benchmark items (questions, ground truth, fault-injected images) are original contributions of this work. The AITEE and CGHD dataset creators are not authors of CircuitBench.

TABLE I
CircuitBench Dataset Composition

Task	Subtask	Dev	Test
A — Component	component_presence	1,158	498
	component_count	975	410
	component_enumeration	579	249
	A total	2,712	1,157
B — Topology	topology_structure	577	249
	node_count	577	249
	component_total	579	249
	B total	1,733	747
E — Fault	fault_presence	859	368
	circuit_description	433	190
	fault_diagnosis	280	119
	E total	1,572	677
Total		6,017	2,581

Task E includes 798 fault-injected items from 200 source circuits (up to two fault types each, yielding E1 and E2 items per fault).

TABLE II
Evaluated Models

Model	Parameters	Quant.
Gemma 3 4B [13]	4.3B	Q4_K_M
LLaVA 7B [10]	7B	Q4_K_M
LLaVA-Phi3 [10], [14]	4B	Q4_K_M

All models accessed via Ollama on an Intel Core Ultra CPU with 16 GB RAM. Model sizes: Gemma 3 4B (3.3 GB), LLaVA 7B (4.7 GB), LLaVA-Phi3 (2.9 GB).

IV. Experimental Setup

A. Evaluated Models

Table II lists the three models evaluated. All are open-weight models run locally via Ollama using 4-bit quantized GGUF weights, requiring no API keys.

B. Prompting Protocol

Each item is presented as a single user turn: the circuit image followed by a task-specific prompt template. Templates are identical across all models; no system prompt is used. Example templates:

Fault presence (E1): "Analyze the electrical circuit shown in the image. Is there a circuit fault present? Answer Yes or No."

Topology structure (B2): "Examine the electrical circuit diagram carefully. Pay attention to how the components are connected. {question} Respond with only 'Yes' or 'No'. Do not include any explanation."

Component count (A2): "Examine the electrical circuit diagram in the provided image. {question} Respond with only the number. Do not include any explanation."

All prompt templates are included in the benchmark release.

C. Inference Parameters

Temperature: 0; maximum output tokens: 256; no in-context examples (zero-shot). A 0.5-second delay is applied between requests.

TABLE III
Overall and Per-Task Accuracy (Development Split)

Model	Overall	Task A	Task B	Task E
Gemma 3 4B	38.3%	38.6%	29.2%	53.8%
LLaVA 7B	23.3%	26.7%	33.3%	0.0%
LLaVA-Phi3	8.0%	15.4%	0.0%	0.0%
n (items)		81 / 30 / 50	81 / 30 / 50	81 / 30 / 50

Model order: Gemma 3 4B / LLaVA 7B / LLaVA-Phi3.

D. Evaluation Metrics

Binary tasks (E1, B2, A1): accuracy based on whether the first token of the response matches the ground-truth "yes" or "no" (case-insensitive).

Integer tasks (A2, B1): accuracy based on whether the first integer token in the response matches the ground-truth count.

Free-text tasks (E2, A3, E3): keyword matching over component names and counts from the ground truth. A response is marked correct only if all required keywords are present. This scorer is conservative and may undercount genuinely correct responses phrased differently.

E. Evaluation Sample

The evaluation presented here uses a development-split sample: 81 items for Gemma 3 4B, 50 items for LLaVA-Phi3, and 31 items for LLaVA 7B, drawn from the 6,017-item development split. All items were sampled before model evaluation; no test-split items were used. The 2,581-item frozen test split is reserved for future full-scale evaluation.

V. Results

A. Overall Performance

Table III reports overall and per-task accuracy for all three models. Gemma 3 4B achieves the highest overall accuracy at 38.3%, followed by LLaVA 7B at 23.3%, and LLaVA-Phi3 at 8.0%. No model exceeds 40% on any task category, and all models perform at or near 0% on Task B (topology/connectivity reasoning) in aggregate—largely driven by complete failure on node-count and component-total subtasks.

B. Performance by Subcategory

Table IV provides a detailed breakdown. Several patterns are immediately apparent.

Binary yes/no tasks are disproportionately easy. Both Gemma 3 4B and LLaVA 7B achieve 87.5% and 100% respectively on topology_structure, and Gemma 3 4B achieves 87.5% on fault_presence. These are binary forced-choice tasks; performance may partly reflect response-format bias rather than genuine circuit understanding.

Counting tasks are inconsistent. LLaVA 7B achieves 50% on component_count while Gemma 3 4B achieves 26.3% and LLaVA-Phi3 0%. However, all three models

TABLE IV
Accuracy by Task Subcategory

Subcategory	Gemma 3 4B	LLaVA 7B	LLaVA-Phi3
Task A — Component Recognition			
component_presence	47.1%	16.7%	40.0%
component_count	26.3%	50.0%	0.0%
component_enumeration	50.0%	0.0%	0.0%
Task B — Topology Reasoning			
topology_structure	87.5%	100.0%	0.0%
node_count	0.0%	0.0%	0.0%
component_total	0.0%	0.0%	0.0%
Task E — Fault Diagnosis			
fault_presence	87.5%	0.0%	0.0%
fault_diagnosis	0.0%	0.0%	0.0%
circuit_description	0.0%	0.0%	0.0%

score 0% on node_count and component_total, suggesting that counting nodes or total components from circuit images is beyond current local VLM capability.

Free-text tasks fail universally. All three models score 0% on fault_diagnosis and circuit_description. This is partly a scorer artifact (strict keyword matching), but manual inspection confirms that models generally either hallucinate component types, describe non-existent circuit features, or produce responses that do not match the expected component-name-plus-count format.

Fault diagnosis (free-text) vs. fault presence (binary). The contrast between Gemma 3 4B's 87.5% on fault_presence and 0% on fault_diagnosis is striking. The model can detect that something is wrong but cannot describe what is wrong in a scorable format.

VI. Error Analysis

Manual inspection of a sample of incorrect responses reveals the following recurring patterns.

Format non-compliance. LLaVA-Phi3 frequently ignores instructions to answer with only "Yes", "No", or a number, instead producing verbose descriptions beginning with phrases like "The circuit shows…" This explains its 0% on topology_structure despite LLaVA 7B (same architecture, larger model) achieving 100%—the smaller Phi3 base model is less instruction-following.

Counting errors. For component_count, models typically answer with a value 1–3 away from the correct count. Gemma 3 4B tends to undercount (e.g., reporting 2 resistors when 4 are present), consistent with failing to distinguish closely placed or overlapping symbols. LLaVA 7B is more accurate on this subtask (50%) but inconsistent across circuit types.

Node-count failure. All models score 0% on node_count. Node counting requires understanding circuit topology globally—tracing wires to identify all distinct electrical nodes—which appears to exceed the spatial reasoning capability of all evaluated models in this zero-shot setting.

Hallucinated fault descriptions. On fault_diagnosis items, Gemma 3 4B typically produces responses describing a circuit fault in general terms (e.g., "open circuit detected") without correctly identifying the specific affected component or location. This is semantically close to correct but fails strict keyword scoring. A rubric-based human evaluation would likely credit partial understanding for these responses.

Component enumeration format mismatch. On circuit_description, all models produce syntactically correct descriptions (e.g., "The circuit contains three resistors and one voltage source") but fail the scorer because the expected format is "Component (N)" notation. These responses represent a scorer limitation as much as a model failure and should be graded manually before drawing conclusions about this subcategory.

VII. Discussion

The results reveal a clear capability hierarchy among the three evaluated models (Gemma 3 4B > LLaVA 7B > LLaVA-Phi3) and a consistent pattern across all models: binary classification tasks are handled far better than counting or free-text generation tasks.

The gap between fault_presence (87.5% for Gemma 3 4B) and fault_diagnosis (0%) is the most practically significant finding. A system that detects a fault but cannot explain it provides limited value to a student debugging a circuit. This suggests that current small open-weight VLMs may be usable as a rough "something looks wrong" detector but not as a reliable diagnostic tool.

The universal failure on node_count (0% across all models) points to a fundamental gap in circuit-graph understanding. Counting nodes requires tracing wire connectivity across the image—a form of spatial relational reasoning that appears absent in all three models tested. This is particularly relevant for AI tutoring applications where Kirchhoff's current law analysis requires correct node identification.

LLaVA-Phi3's near-complete failure (8% overall, 0% on all tasks except component_presence) appears primarily attributable to instruction-following breakdown: the 4B Phi3 base model frequently ignores format constraints, producing verbose answers where single tokens are expected. This is a practical deployment concern independent of circuit understanding.

The benchmark's current evaluation uses development-split samples of 30–81 items per model, which is sufficient to identify broad patterns but insufficient for fine-grained statistical comparisons between models. The frozen 2,581-item test split is reserved for a future evaluation that includes cloud-API models (GPT-4o, Gemini, Claude) for comparison against the open-weight baselines established here.

VIII. Limitations

1) Small evaluation sample. Development-split samples of 30–81 items per model support pattern identification but not rigorous statistical model comparison.

Full test-split evaluation is needed for final conclusions.

2) Open-weight models only. The evaluated models are small, locally-hosted, 4-bit quantized. State-of-the-art cloud-API VLMs (GPT-4o, Gemini 2.0 Flash, Claude Sonnet) may perform substantially better.
3) Conservative free-text scorer. Keyword matching undercounts partially correct free-text responses. Subcategories including fault_diagnosis and circuit_description require human review to calibrate.
4) Benchmark source. CircuitBench is derived from AITEE and CGHD, which were not designed as VLM evaluation resources. The circuit distribution reflects what is available in these datasets.
5) Circuit complexity. AITEE depicts first-semester educational circuits. Results may not generalize to more complex topologies.
6) Model contamination. AITEE and CGHD are publicly available. Contemporary VLMs may have encountered their images during pretraining, and high accuracy should not be interpreted as pure unseen generalization.
7) Synthetic faults. The two fault types (open circuit, disconnected component) are controlled synthetic modifications. Real student wiring mistakes may exhibit different visual characteristics.
8) Zero-shot prompting only. Performance may improve with few-shot examples or chain-of-thought prompting; this is not explored here.

IX. Conclusion

We introduced CircuitBench, a source-backed benchmark of 8,598 items for evaluating VLMs on electronic-circuit understanding, reasoning, and fault diagnosis, derived from the AITEE and CGHD datasets. Evaluation of three locally-hosted open-weight VLMs reveals a consistent pattern: binary yes/no tasks (fault presence, topology structure) are handled reasonably well by the best model (Gemma 3 4B: 87.5%), while counting tasks, free-text diagnosis, and node identification fail near-universally across all models. Overall accuracy ranges from 8.0% (LLaVA-Phi3) to 38.3% (Gemma 3 4B)—well below the reliability threshold required for autonomous instructional use.

These results suggest that small open-weight VLMs in their current form are not suitable as autonomous circuit-debugging assistants, though they may be useful as coarse anomaly detectors or component-presence checkers in human-in-the-loop workflows.

Future work includes full test-split evaluation with cloud-API models, extending the benchmark to hand-drawn circuit representations (Task F), adding quantitative reasoning tasks (Task D), and calibrating the free-text scorer against human judgments.

Data and Code Availability

The CircuitBench benchmark manifest (JSONL), evaluation scripts, scored results, and fault-injected images are available at https://github.com/sasha-sam/circuitbench. Source datasets: AITEE [1] and CGHD [2].

AI Assistance Disclosure

Generative AI tools, including Claude (Anthropic), were used to assist with code development, manuscript organization, and language refinement during preparation of this work. All research questions, experimental design choices, data-source selection, source verification, model evaluations, result interpretation, citation verification, and final manuscript content were reviewed and validated by the author. AI systems were not treated as authors.

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